

MA 214 Week 2: Lab 2

Regression Models

1. **Your name:** _____

BU ID: _____

2. **Partner's name:** _____

BU ID: _____

Big idea: Regression models describe how a response variable changes with one or more predictors, but the model must be interpreted and checked in context. **Do not use GenAI tools for this lab.** The point is to practice your own analysis work, coding skills, and statistical reasoning.

Files used: `mtcars.csv`, `lab2_data.csv`, `lab2_data2.csv`, and `lab-starter.R`

Skills practiced: `lm()`, `summary()`, `cor()`, residual plots, and R^2 comparisons.

Your mission: Fit and compare regression models that predict fuel efficiency, then explain which model is most useful and why.

Icebreaker

Before opening R: introduce yourself to your partner and answer the warm-up question below.

Warm-up response: Name one variable that might predict a car's miles per gallon. Why do you expect it to matter?

Getting Started

Open `lab-starter.R`. Import `mtcars.csv` and inspect the response and candidate predictors.

```
library(dplyr)
library(ggplot2)

mtcars_df <- read.csv("mtcars.csv")

head(mtcars_df)
str(mtcars_df)
```

HUDDLE UP

Which variable should be the response? Which predictors do you expect to be most strongly related to fuel efficiency?

Question 1. Explore the Data

Make a first scatter plot of `mpg` versus `wt`. Then compute the correlation.

```
ggplot(mtcars_df, aes(x = wt, y = mpg)) +  
  geom_point()  
  
cor(mtcars_df$wt, mtcars_df$mpg)
```

Variable or data set	Type or role	What it means
mpg		
wt		
hp or qsec		

Write 2-3 sentences describing the relationship you see before fitting a model.

Question 2. Make a Prediction

Before fitting the model, predict the sign of the slope for `mpg` predicted by `wt`.

PREDICTION TIME

Will the slope be positive or negative? What would that sign mean in context?

```
model_wt <- lm(mpg ~ wt, data = mtcars_df)  
summary(model_wt)
```

What did your team predict? What evidence did you check first?

Question 3. Compute and Compare Models

Fit two more simple regression models and compare them to `model_wt`.

```
model_hp <- lm(mpg ~ hp, data = mtcars_df)
model_qsec <- lm(mpg ~ qsec, data = mtcars_df)

summary(model_wt)$r.squared
summary(model_hp)$r.squared
summary(model_qsec)$r.squared
```

HUDDLE UP

Does the model with the largest R^2 automatically tell the best statistical story? What else should you check?

CLAIM, EVIDENCE, CONTEXT

The strongest simple model is _____.

My evidence is _____.

This matters because _____.

Question 4. Interpret in Context

Use the model output and a diagnostic plot to explain what the regression says about cars.

Question	Your answer
What comparison did you make?	
What result is strongest?	
What limitation or caution should readers know?	

DEBATE CORNER

If you and your partner disagree about the best model, write both arguments first. Then decide what claim the evidence actually supports.

Question 5. Close the Lab

Submit your completed `lab-starter.R`. Your file should include the fitted models and requested autograder objects.

Final response: state your chosen model, cite one piece of evidence, and interpret the result in context.

Optional Challenge

Fit a multiple regression model using at least two predictors and compare it with your best simple regression model.

```
model_multi <- lm(mpg ~ wt + hp, data = mtcars_df)
summary(model_multi)
```